

TITLE & AUTHOR INFORMATION

Dynamic networks of methicillin-resistant *Staphylococcus aureus* in communities drive hospital transmission as revealed by whole-genome sequencing

Scott A. Cohen¹, Massimiliano S. Tagliamonte^{1,2}, Carla Mavian^{1,2}, Mattia Prosperi¹, Marco Salemi^{1,2}, J. Adrian Tyndall¹, J. Glenn Morris¹

¹ Emerging Pathogens Institute, University of Florida; ² Department of Pathology, Immunology, Laboratory Medicine, College of Medicine.

STUDY OVERVIEW

BACKGROUND

- Methicillin-resistant *Staphylococcus aureus* (MRSA) infections remain a leading cause of antibiotic resistant infections in hospitals
- Recent MRSA trends have promoted uncertainty on transmission pathways, particularly among strains in hospital and community settings.
- A simultaneous application of whole-genome sequencing of both community and hospital *S. aureus* isolates will elucidate modern evolutionary pathways.

OBJECTIVE

- To understand MRSA transmission networks, we performed longitudinal epidemiological and phylogenetic analyses on community and hospital-associated MRSA infections in North-Central Florida.

STUDY DESIGN

- Serial cross-sectional community and hospital sampling were performed at University of Florida Health between 2010 and 2019 (**Figure 1**)
- Community Sampling:** From August 2015-January 2017, pediatric and adult patients presenting to ED with acute skin and soft tissue infection (SSTI) were prospectively enrolled and surveyed
- Hospital Sampling:** MRSA isolates collected from healthcare-acquired infections were sampled before (2010), concurrently, and after (2019) community sampling study period.
- Whole-genome sequencing (Illumina) was conducted on cultured isolates

STATISTICAL ANALYSIS

- Fisher's Exact test and Wilcoxon rank-sum were used to test associations between geographic, medical, and social determinants with microbiological and molecular results among community isolates. (**Table 1**)
- Kernel density models were developed to illustrate the spatial distribution of the community-based MRSA and MSSA SSTI incidence (**Figure 2**)
- Genotype of isolates were characterized via *spa*-typing (**Figure 3**)
- Maximum likelihood tree was calculated using single nucleotide polymorphisms detected in WGS (IQ-TREE) (**Figure 4**)

CONCLUSION

- MRSA transmission in hospital settings was introduced from multiple epidemiologically unrecognized MRSA species with ancestral origins in the community settings.
- The non-uniform geographic distribution, particularly high incidence in rural areas, suggest lifestyle and behavioral patterns may influence transmission in community settings more than traditional genotypic factors.
- Nosocomial MRSA outbreak prevention strategies should target unique aspects of the community rather than focusing solely on the hospital

In North-Central Florida, MRSA found in communities drive transmission of healthcare-acquired MRSA infections in hospital settings



Florida Chapter
American College of Physicians
Spring 2023

RESULTS OVERVIEW: TABLES & FIGURES

Figure 1: Study inclusion flow and sample selection

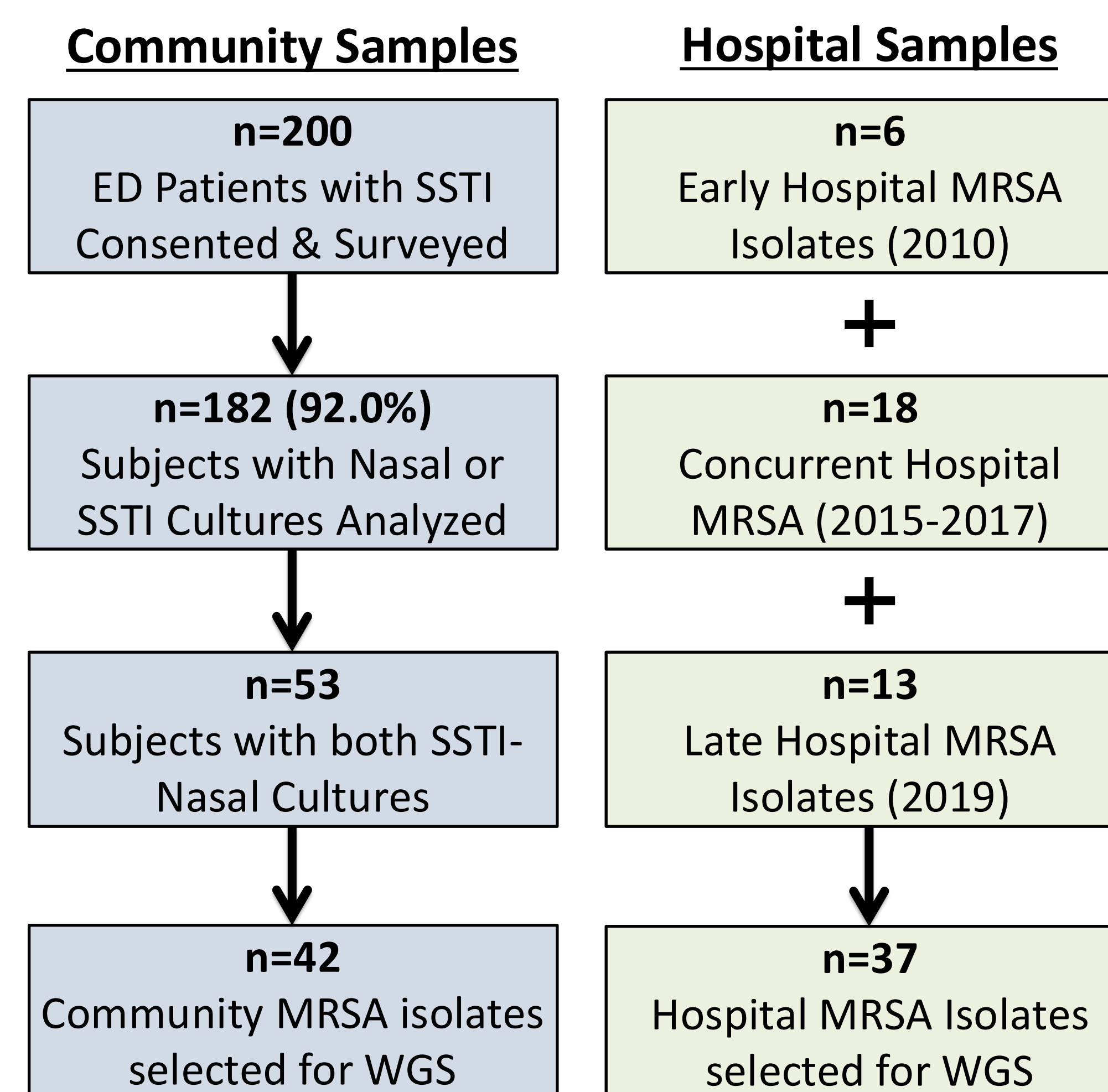


Table 1: Demographics and social determinants by MRSA microbiology of community samples

	SSTI Culture		P Value
	MRSA (n=53)	Non-MRSA (n=129)	
Age:			
Average Age (± SD)	36.4 ± 21.0	37.8 ± 20.3	0.676
Sex:			
Male	31 (58.5)	66 (51.2)	0.415
Female	22 (41.5)	63 (48.8)	
Race:			
African-American	10 (18.9)	31 (24.0)	0.670
White	39 (73.6)	83 (64.3)	
Other	4 (7.5)	15 (11.7)	
Ethnicity			
Hispanic	6 (11.8)	14 (11.0)	0.920
Social History:			
Insurance Coverage	31 (58.5)	91 (70.5)	0.195
Current Alcohol Hx	21 (39.6)	44 (34.1)	0.500
Current Smoke Hx	24 (45.3)	47 (36.4)	0.316
Livestock Exposure	14 (26.4)	14 (10.9)	0.009
Antimicrobial Soap	34 (64.1)	79 (61.7)	0.629
Currently in School	6 (11.3)	23 (18.0)	0.373
Population Density			
<100 Per Sq. Mile	30 (56.6)	48 (37.2)	0.021
≥100 Per Sq. Mile	23 (43.4)	81 (62.8)	
Rural-Urban Status			
Rural	14 (26.4)	17 (13.1)	0.049
Urban	39 (73.6)	112 (86.9)	

Figure 2: Kernel density models* illustrating spatial patterns of (A) all eligible subjects (B) MRSA SSTI (C) MSSA SSTI incidence by self-reported residence.
*Adjusted by neighborhood ED utilization rates

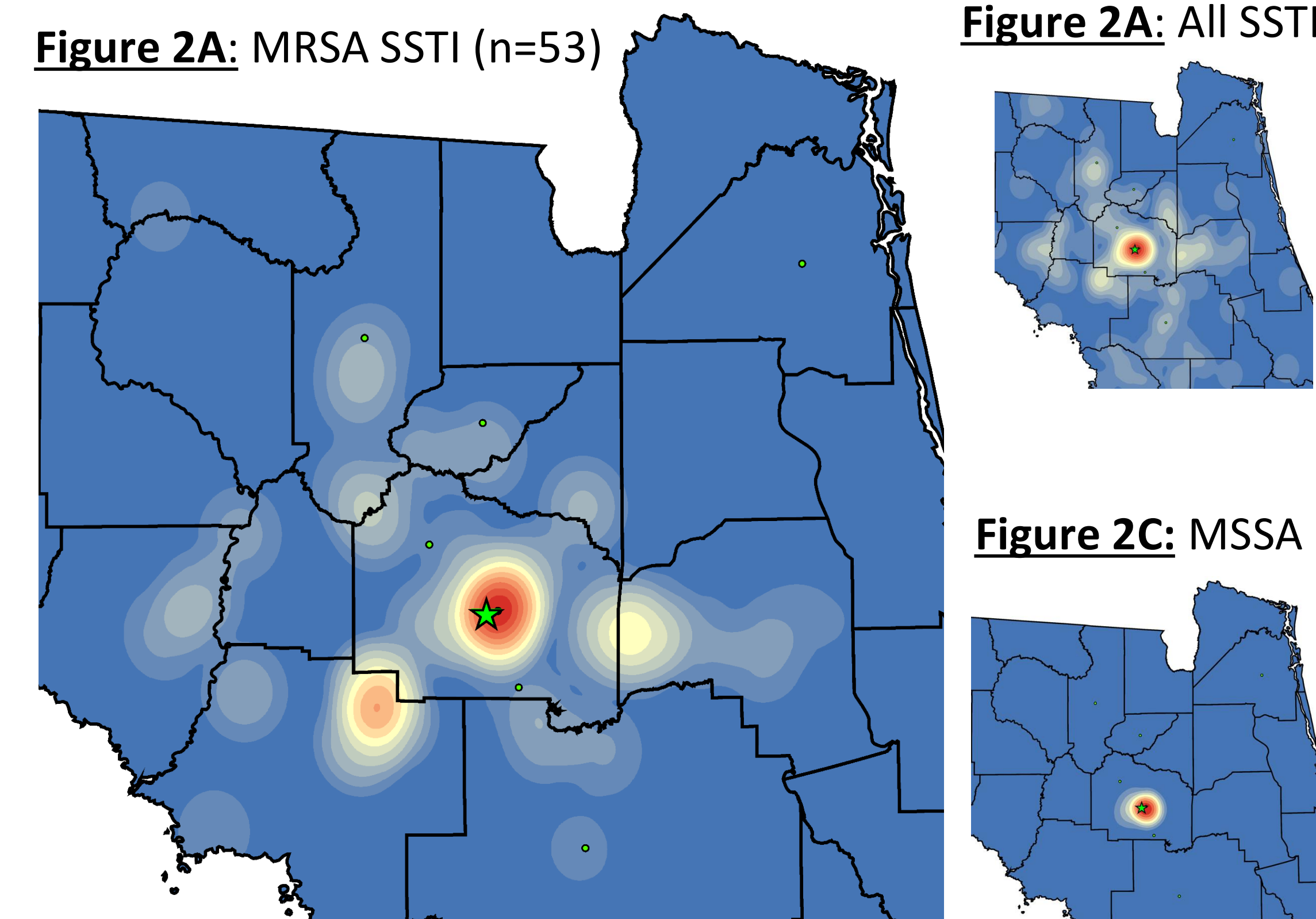


Figure 3: Frequency of *spa*-types by sample source

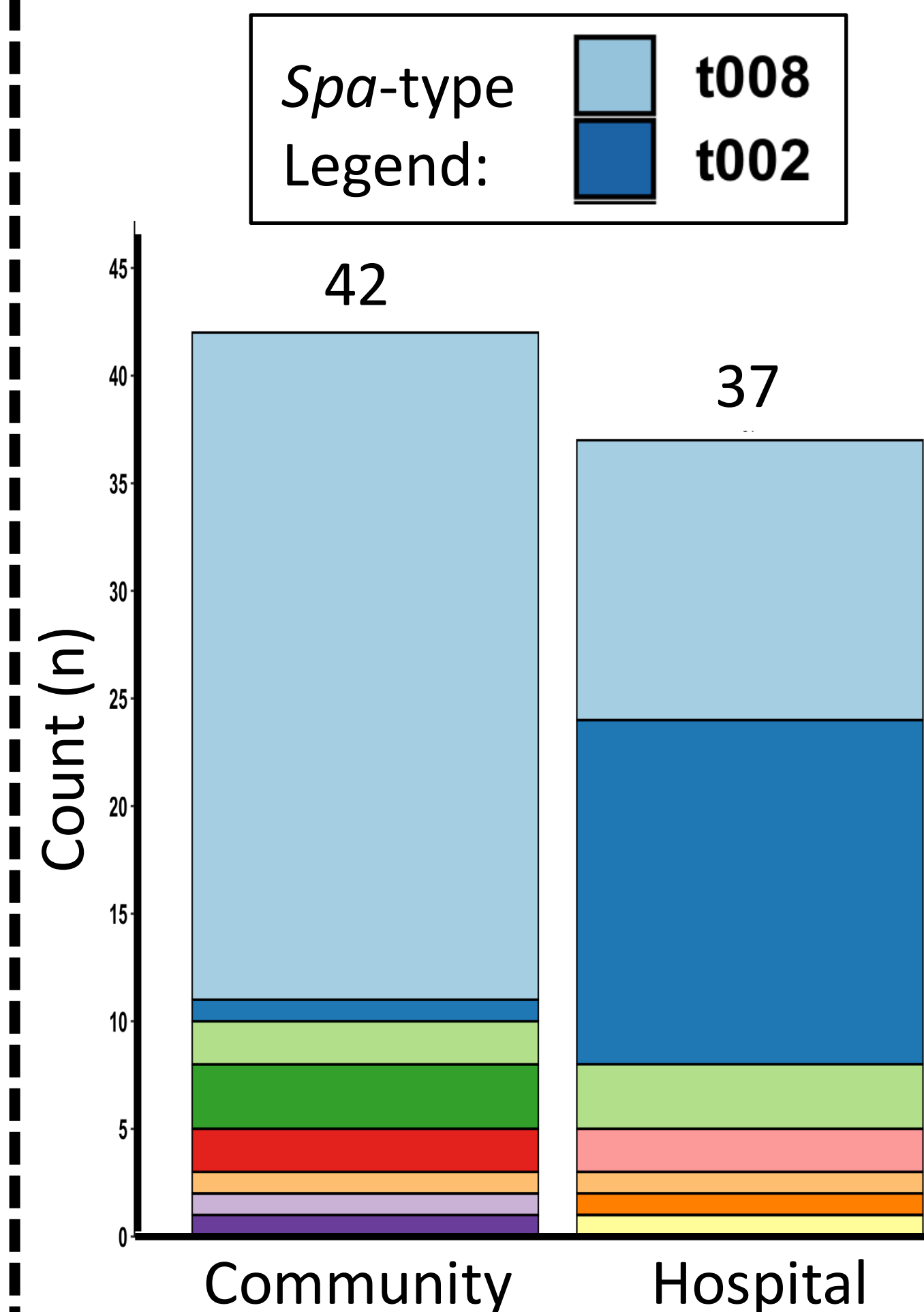


Figure 4: Histogram containing summary of two distinct phylogenetic clades: Large Clade (n=57) and Small Clade (n=22) [Phylogenetic tree of Large Clade is illustrated]

